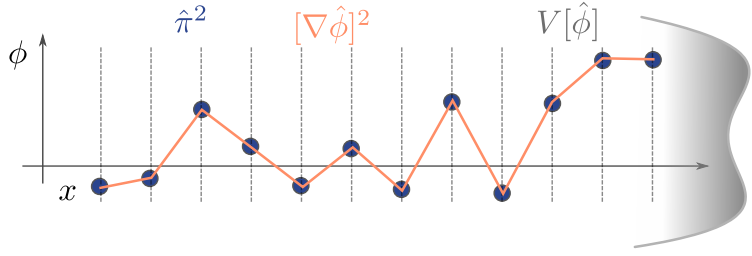
Variational method in quantum field theory

without cutoff

Antoine Tilloy
Max Planck Institute of Quantum Optics, Garching, Germany



University of Vienna
May 12th, 2021

MPQ

Quantum field theory

- Currently a good candidate for the most fundamental theory
- Difficult to define (progress in low dimensions)
- Very difficult to solve non-perturbatively
- A challenge for all numerical methods and tensor networks!

Quantum field theory: different approaches

Two ways to attack *real world* quantum field theories non-perturbatively

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ϕ_2^4 - pile of dirt



QCD - Everest



$\mathcal{N} = 4$ *SYM* - Chrysler building

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Goal - ideal - philosophy: an apology of the pile of dirt approach

Abandon analytical solutions, but find robust methods that can solve simple QFTs non-perturbatively and, if possible, to machine precision, *without cheating*.

more on this on tilloy.wordpress.com

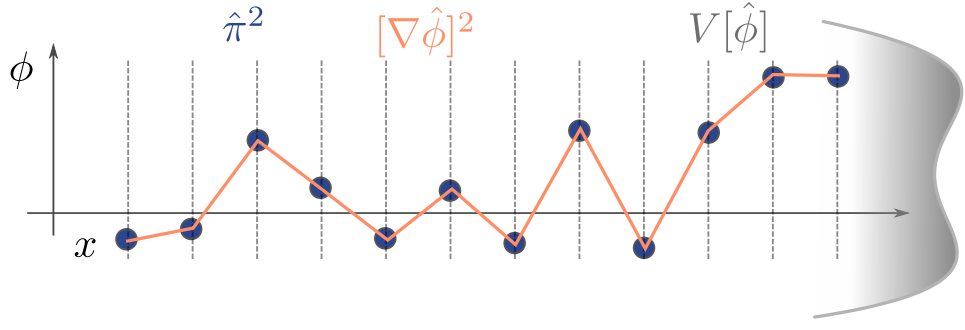
Outline

1. ϕ_2^4 for beginners
2. Some solutions with discretization
3. The variational method
4. Continuous matrix product states
5. Going relativistic
6. Results and improvements

ϕ_2^4 for beginners

and condensed matter theorists

Intuitive definition: canonical quantization



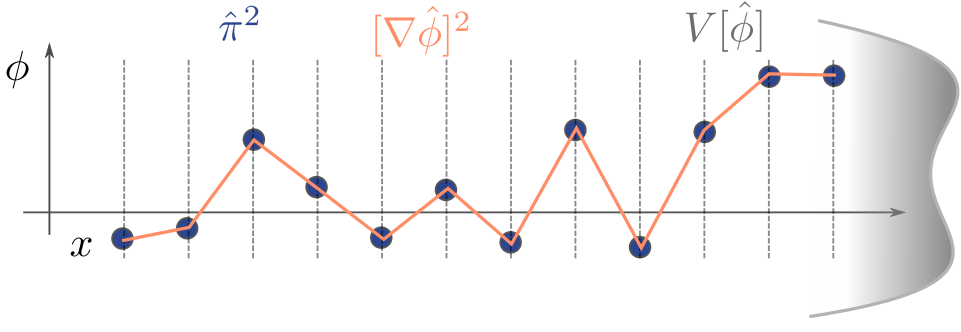
Hamiltonian

A continuum of nearest neighbor coupled anharmonic oscillators

$$\hat{H} = \int_{\mathbb{R}^d} \mathrm{d}^d x \quad \underbrace{\frac{\hat{\pi}(x)^2}{2}}_{\text{on-site inertia}} + \underbrace{\frac{[\nabla\hat{\phi}(x)]^2}{2}}_{\text{spatial stiffness}} + \underbrace{V(\hat{\phi}(x))}_{\text{on-site potential}}$$

with canonical commutation relations $[\hat{\phi}(x), \hat{\pi}(y)] = i\delta^d(x-y)\mathbb{1}$ (*i.e.* bosons)

Intuitive definition



Hilbert space

Fock space $\mathscr{H}_{\text{QFT}} = \mathscr{F}[L^2(\mathbb{R}^d)]$ – just like $x, p \to (a, a^\dagger)$ do $\hat{\pi}, \hat{\phi} \to \hat{\psi}, \hat{\psi}^\dagger$

$$|\Psi\rangle = \sum_{n=0}^{+\infty} \int \mathrm{d}x_1 \mathrm{d}x_2 \cdots \mathrm{d}x_n \underbrace{\varphi_n(x_1, x_2, \cdots, x_n)}_{\text{wave function}} \underbrace{\hat{\psi}^\dagger(x_1)\hat{\psi}^\dagger(x_2)\cdots\hat{\psi}^\dagger(x_n)}_{\text{local oscillator creation}} |\text{vac}\rangle$$

Why relativistic? $\rightarrow$ functional integral

Insert $\mathbb{1} = \int \mathcal{D}\phi \, |\phi\rangle\langle\phi|$ in expression for correlation functions and $t = i\tau$ gives

Functional integral representation

Representation of correlation functions in terms of random fields

$$\langle 0|\hat{\phi}(\tau_1, x_1) \cdots \hat{\phi}(\tau_n, x_n)|0\rangle := \int \phi(\tau_1, x_1) \cdots \phi(\tau_1, x_n) \; e^{-S(\phi)} \underbrace{\mathcal{D}\phi}_{\text{“Lebesgue measure”}}$$

with the action / weight where $\hat{\pi} \rightarrow \frac{\mathrm{d}\phi}{\mathrm{d}\tau}$

$$S(\phi) = \int \mathrm{d}^d x \, \mathrm{d}\tau \quad \underbrace{\frac{1}{2}\left[\frac{\mathrm{d}\phi}{\mathrm{d}\tau}\right]^2}_{\text{inertia a.k.a time stiffness}} + \underbrace{\frac{[\nabla\phi]^2}{2}}_{\text{spatial stiffness}} + \underbrace{V(\phi)}_{\text{on-site potential}}$$

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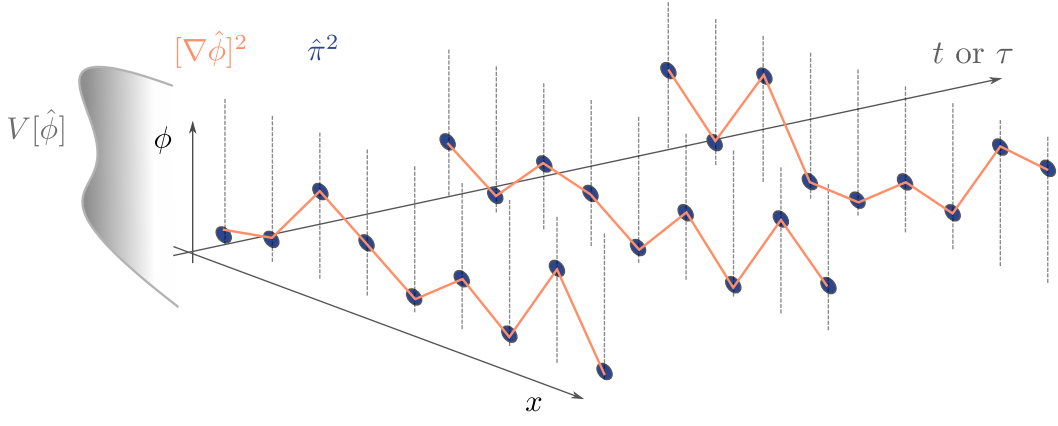
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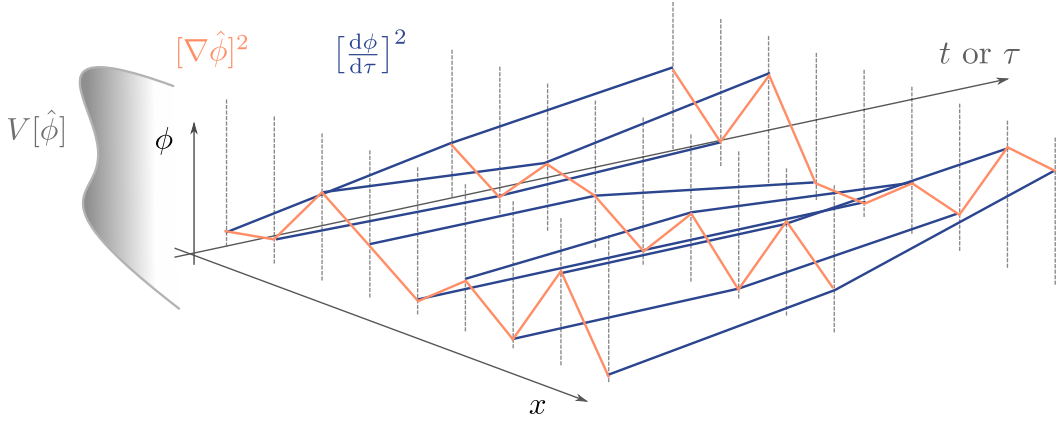
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Inertia = time stiffness $\Longrightarrow$ Euclidean rotation invariance $\Longrightarrow$ Lorentz

Why relativistic? $\rightarrow$ functional integral



Why relativistic? $\rightarrow$ functional integral



What are the problems - Hilbert space approach

The Hamiltonian is ill defined on all states in the Hilbert space because of infinite zero point energy *i.e.* terms $\propto \hat{\psi}(x)\hat{\psi}^{\dagger}(x)$

$$\langle \Psi_1 | \hat{H} | \Psi_2 \rangle = \pm\infty \quad \text{and even} \quad \langle \text{vac} | \hat{H} | \text{vac} \rangle \propto \delta^d(0) = +\infty$$

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If the divergent vacuum terms are removed, the Hamiltonian is not bounded from below

$$\forall\ |\Psi\rangle \in \mathscr{H},\ \langle \Psi | \hat{H}_{\text{finite}} | \Psi \rangle = \text{finite but}\ \ \exists\ \Psi_n \text{ s.t. } \lim_{n \to +\infty} \langle \Psi_n | H_{\text{finite}} | \Psi_n \rangle = -\infty$$

How are they are solved in the free case - Hamiltonian

Bogoliubov transform

Go from $\hat{\psi}(x), \hat{\psi}^\dagger(x)$ to $a(p), a^\dagger(p)$ with

$$a(p) = \frac{1}{\sqrt{2}} \left(\sqrt{\omega_p}\, \hat{\phi}(p) + i \frac{\hat{\pi}(p)}{\sqrt{\omega_p}} \right) \quad \text{with} \quad \omega_p = \sqrt{p^2 + m^2}$$

which yields

$$H_0 = \int \mathrm{d}p\, \omega_p\, \frac{1}{2} \left(a_p^\dagger a_p + a_p a_p^\dagger \right)$$

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Solution

- Take $H_{\mathrm{QFT}} \equiv : H :_a$
- $|\text{free ground state}\rangle = |\text{vacuum}\rangle_a$
- $\mathscr{H}$ built from $a_{p_1}^\dagger \cdots a_{p_n}^\dagger |\text{vacuum}\rangle_a$

This solves the problematic free part exactly, and allows to define a finite interaction (in $1+1$)

Rigorous operator definition of ϕ_2^4

Renormalized ϕ_2^4 theory

$$H = \int \mathrm{d}x \; \frac{:\pi^2:_a}{2} + \frac{:(\nabla\phi)^2:_a}{2} + \frac{m^2}{2} :\phi^2:_a + g :\phi^4:_a$$

(note that $:\diamondsuit:_a$ depends on m)

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(note that $:\diamondsuit:_a$ depends on m)

1. Rigorously defined relativistic QFT without cutoff (Wightman QFT)
2. Vacuum energy density finite
3. Very difficult to solve unless $g \ll m^2$ (perturbation theory)
4. Phase transition around $f_c = \frac{g}{4m^2} = 11$ i.e. $g \simeq 2.7$ in mass units

Ways to solve the non-exactly-solvable

The two main games in town

1. Perturbative expansions (+ Borel-Padé resummation)
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2. Non-perturbative renormalization group (Kadanoff, FRG, Tensor RG, etc.)

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The two new methods now rule on (low dimensional) lattice problems thanks to tensor networks $\rightarrow$ QFT?

Solving on the lattice

the state of the art

lattice ϕ_2^4

First non-trivial example to have been rigorously constructed this way

$$S(\phi) = \sum_{\langle i,j \rangle} \underbrace{\frac{(\phi_i - \phi_j)^2}{2a^2}}_{(\nabla\phi)^2/2} a^2 + \sum_i \frac{1}{2}\mu_a^2 \phi_i^2 + \frac{1}{4}\lambda_a \phi_i^4$$

Taking the limit

The right way to get the continuum limit is to take:

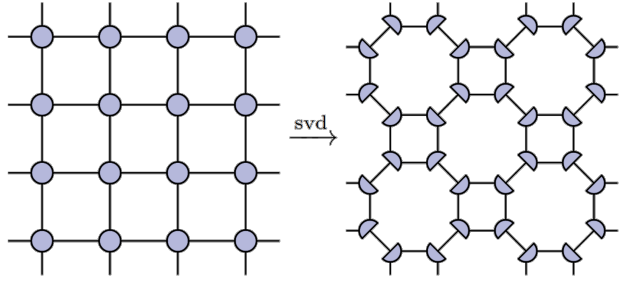
$$\mu_a = \mu a^2 + \frac{3}{2}\log(a) a^2 \lambda$$
$$\lambda_a = \lambda a^2$$

which is equivalent to normal ordering the interaction term.

Basically, at first order in perturbation theory, the ϕ^4 term behaves like a ϕ^2 term times a log divergent constant.

Example with tensor network renormalization

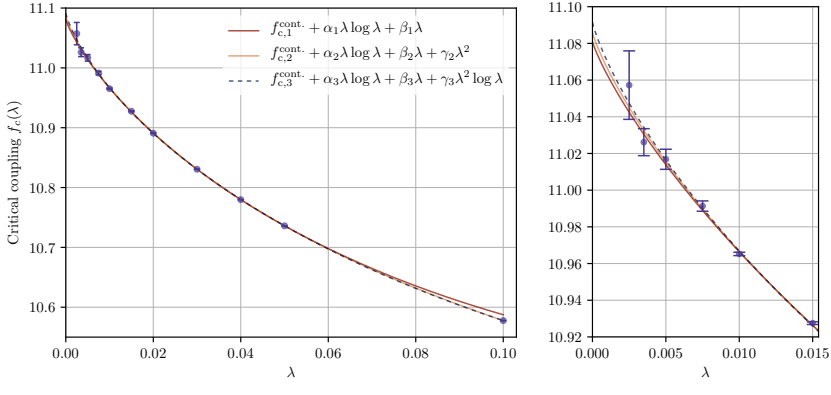
Done with Clément [late 2019 – early 2020]
Discretize ϕ, write $Z = \sum S(\phi)$ as a tensor network and contract it with TRG + GILT



Technically: UV cutoff (lattice) and IR cutoff (number of RG steps)

Example with tensor network renormalization

Continuum limit taken **numerically**



More costly as the UV cutoff gets small because:

1. Field becomes unbounded at short distance $\rightarrow$ large starting bond dimension
2. More RG steps (with max χ) to get to the same scale

Limitation of numerical continuum limit

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$\Longrightarrow$ even theory independent: would apply to QCD (asymptotic freedom)

Results

For ϕ_2^4, critical coupling $f_c = \lambda/\mu^2$

Method	$f_c^{\text{cont.}}$	Year	Ref.
Tensor network coarse-graining	10.913(56)	2019	[9]
Borel resummation	11.23(14)	2018	[6]
Renormalized Hamil. Trunc.	11.04(12)	2017	[5]
Matrix Product States	11.064(20)	2013	[7]
Monte Carlo	11.055(20)	2019	[15]
This work	11.0861(90)	2020	

TABLE I. Comparison of several estimates of the critical coupling constant $f_c^{\text{cont.}}$ in the continuum obtained using different methods.

New results fresh from Ghent with MPS + finite entanglement scaling + continuum limit scaling $f_c = 11.09698(31)$ [arXiv:2104.10564]

see tilloy.wordpress.com for a discussion

Why bother? True vs Effective QFT

Against the "why bother since there is always a cutoff?"

Effective QFT

The theory has a momentum/energy cutoff Λ large but finite $\Lambda \gg m$, where m is the gap.

The fundamental theory is not known, but in perturbation theory, one can take $\Lambda \to \infty$ term by term to get a good approximation of physics at scale m.

Examples

1. QED with matter (probably..)
2. ϕ_4^4

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True QFT

The limit $\Lambda \to +\infty$ can be taken exactly, and the theory is valid "all the way down".

All quantities exist non-perturbatively in the limiting theory, for arbitrarily high energy. No cutoff whatsoever in principle.

Examples

1. QCD without too much matter
2. ϕ_2^4 and ϕ_3^4
3. Sine-Gordon, Gross-Neveu, etc.

The variational method

in the continuum

The variational method

In the Hamiltonian formulation:

- Guess a **finite dimensional submanifold** $\mathscr{M}$ of the QFT Hilbert space $\mathscr{H}$
- Find the ground state by minimizing $\langle H \rangle$:

$$|\text{ground}\rangle \simeq |\psi\rangle = \underset{\mathscr{M}}{\text{argmin}} \frac{\langle \psi | H | \psi \rangle}{\langle \psi | \psi \rangle}$$

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Example: naive Hamiltonian truncation

With an IR cutoff, momenta are discrete. Take as submanifold $\mathscr{M}$ the **vector space** spanned by:

$$a^\dagger_{k_1} a^\dagger_{k_2} \cdots a^\dagger_{k_r} |0\rangle_a$$

where $r \leqslant r_{\text{max}}$ and $k \leqslant k_{\text{max}}$ (one possible truncation)

Feynman's objection

Feynman's requirement for variational wavefunctions in RQFT (1987)

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All methods so far break one at least:

- Hamiltonian truncation fails at 1 (but works fairly well through its renormalized refinements)
- Tensor networks succeed at 1 and 2 but fail (a priori) at 3

Haegeman-Cirac-Osborne-Verschelde-Verstraete fix of 2010: regulate the UV by adding a Lagrange multiplier in the Hamiltonian $H \to H + \frac{1}{\Lambda^2}\,\text{regulator}$

(Continuous) matrix product states

Taking the simplest tensor network and scaling it up to QFT

Continuous Matrix Product States

Introduced by Verstraete and Cirac in 2010

Definition

$$|Q,R\rangle = \mathrm{tr}\left[\mathcal{P}\exp\left\{\int_0^L dx\; Q\otimes\mathbb{1} + R\otimes\psi^\dagger(x)\right\}\right]|0\rangle_\psi$$

- Q, R are $D\times D$ matrices,
- The trace is taken over this matrix space
- $[\psi(x),\psi^\dagger(y)] = \delta(x-y)$
- $\psi^\dagger(x)$ is non-relativistic creation operator (i.e. $\phi(x) = \frac{1}{\sqrt{2\nu}}[\psi(x)+\psi^\dagger(x)]$)
- $|0\rangle_\psi$ is the associated Fock vacuum

Idea:

- From MPS: a continuum limit
- From QFT: a sort of generalized "non-commutative" coherent state

Computations

Some correlation functions

$$\left\langle \hat{\psi}(x)^\dagger \hat{\psi}(x) \right\rangle = \mathrm{Tr}\left[e^{TL} (R \otimes \overline{R}) \right]$$

$$\left\langle \hat{\psi}(x)^\dagger \hat{\psi}(0)^\dagger \hat{\psi}(0) \hat{\psi}(x) \right\rangle = \mathrm{Tr}\left[e^{T(L-x)} (R \otimes \overline{R}) e^{Tx} (R \otimes \overline{R}) \right]$$

$$\left\langle \hat{\psi}(x)^\dagger \left[-\frac{d^2}{dx^2} \right] \hat{\psi}(x) \right\rangle = \mathrm{Tr}\left[e^{TL} ([Q, R] \otimes [\overline{Q}, \overline{R}]) \right]$$

with $T = Q \otimes \mathbb{1} + \mathbb{1} \otimes \bar{Q} + R \otimes \bar{R}$

Example

Lieb-Liniger Hamiltonian

$$\mathcal{H} = \int_{-\infty}^{+\infty} dx \left[\frac{d\hat{\psi}^\dagger}{dx} \frac{d\hat{\psi}}{dx} - \mu \hat{\psi}^\dagger \hat{\psi} + c \hat{\psi}^\dagger \hat{\psi}^\dagger \hat{\psi} \hat{\psi} \right]$$

Solve by **minimizing**: $\langle Q, R | \mathcal{H} | Q, R \rangle = f(Q, R)$

Standard CMPS and ϕ^4

Applying cMPS to the ϕ^4 Hamiltonian

$$\langle Q, R|\hat{h}_{\phi^4}|Q, R\rangle = \infty$$

Oh no!

The short distance behavior of cMPS is the wrong one, even the free theory is hard to approximate.

A possible fix by Haegeman-Cirac-Osborne-Verschelde-Verstraete:

$$H \to H_\Lambda := H + \frac{1}{\Lambda^2}\int \mathrm{d}x\, \frac{(\partial_x \pi)^2}{2}$$

Going relativistic

Infusing some “high-energy” knowledge into tensor networks

Towards relativistic CMPS

Local basis in position of the QFT: $\psi^\dagger$, ϕ, π, $|0\rangle_\psi$
Diagonal basis of the free part: $a_k^\dagger$, $|0\rangle_a$

Bogoliubov transform

Go from $\hat{\psi}(x), \hat{\psi}^\dagger(x)$ to $a(p), a^\dagger(p)$ with

$$a(p) = \frac{1}{\sqrt{2}}\left(\sqrt{\omega_p}\,\hat{\phi}(p) + i\frac{\hat{\pi}(p)}{\sqrt{\omega_p}}\right) \quad \text{with} \quad \omega_p = \sqrt{p^2 + m^2}$$

which yields

$$H_0 = \int \mathrm{d}p\, \omega_p\, \frac{1}{2}\left(a_p^\dagger a_p + a_p a_p^\dagger\right)$$

Go from $|0\rangle_\psi$ to $|0\rangle_a$
and
Go from $\psi(x)$ to $a(x) = \int \mathrm{d}p\; a(p)\mathrm{e}^{ipx} \neq \psi(x)$

Relativistic CMPS

Definition

$$|R, Q\rangle = \mathrm{tr}\left\{\mathcal{P}\exp\left[\int \mathrm{d}x\, Q \otimes \mathbb{1} + R \otimes a^\dagger(x)\right]\right\}|0\rangle_a$$

Some properties

1. $|0,0\rangle = |0\rangle_a$ is the ground state of H_0 hence exact CFT UV fixed point (because interaction super-renormalizable)
2. $\langle Q, R|h_{\phi^4}|Q, R\rangle$ is finite for all Q, R (not trivial)

Consequence on the Hamiltonian

Hamiltonian density in $a(x)$ basis

H is local in $\psi(x)$, not in $a(x)$...

$$H = \int \mathrm{d}x_1 \mathrm{d}x_2 D(x_1 - x_2) a^\dagger(x_1) a(x_2)$$
$$+ \int \mathrm{d}x_1 \mathrm{d}x_2 \mathrm{d}x_3 \mathrm{d}x_4 K(x_1, x_2, x_3, x_4) a(x_1) a(x_2) a(x_3) a(x_4) + 4 a^\dagger aaa + 3 a^\dagger a^\dagger aa$$
$$+ \text{h.c.}$$

But fortunately exponentially decreasing: K is horrible, but decays $\propto e^{-m|x|}$.

The nightmarish optimization

Procedure:
Compute $e_0 = \langle Q, R | h_{\phi^4} | Q, R \rangle$ and $\nabla_{Q,R} e_0$ and minimize

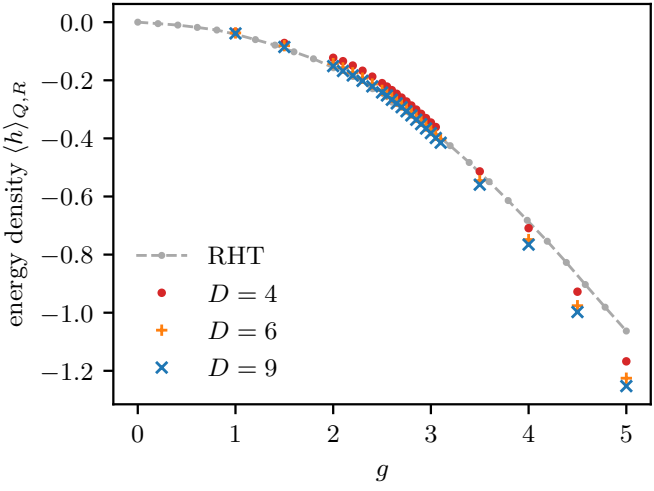
Computations in a nutshell:

1. Contains an algebraic part identical to standard cMPS
2. Involves horrible quadruple integrals without analytic solutions
   NEW: can be calculated efficiently $\propto D^3$ with linear ODE

Optimization a priori non-trivial but **efficient** with geometric methods (gradient descent on a manifold with a natural metric aka **imaginary time TDVP**)

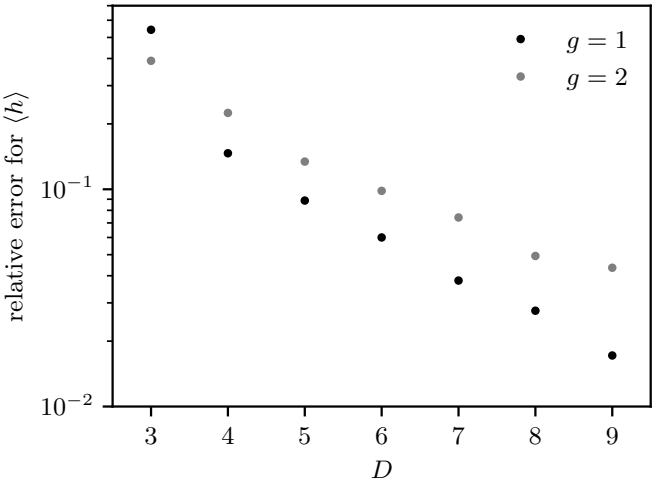
Results and discussion

Results



Compared with the Renormalized Hamiltonian Truncation results of Rychkov and Vitale from 2015. **NEW**: up to $D = 20$

Results



Compared with the "high precision" Renormalized Hamiltonian Truncation results of Elias Miro, Rychkov, and Vitale from 2017 for $g = 1$ and $g = 2$.
NEW: up to $D = 20$, better than RHT, error $< 10^{-3}$

Comparison with renormalized Hamiltonian truncation

Ren. Hamiltonian truncation

IR cutoff L, energy truncation E_T

- Uses a vector space
- Function to minimize is quadratic, hence linear problem
- Number of parameters $\propto \mathrm{e}^{L \times E_T}$
- Error $\propto 1/E_T^3$
- Spectrum easy

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Relativistic CMPS

entanglement truncation D

- Uses a manifold
- Minimization is a priori non-trivial but doable
- Number of parameters $\propto D^2$
- Error $o(1/D^\alpha)$, $\forall\ \alpha$ (folklore)
- Fixed t correl. functions easy

Comparison with renormalized Hamiltonian truncation

Ren. Hamiltonian truncation

IR cutoff L, energy truncation E_T

- Uses a vector space
- Function to minimize is quadratic, hence linear problem
- Number of parameters $\propto \mathrm{e}^{L \times E_T}$
- Error $\propto 1/E_T^3$
- Spectrum easy

Relativistic CMPS

entanglement truncation D

- Uses a manifold
- Minimization is a priori non-trivial but doable
- Number of parameters $\propto D^2$
- Error $o(1/D^\alpha)$, $\forall\ \alpha$ (folklore)
- Fixed t correl. functions easy

Note: real world not asymptotic. RCMPS has expensive prefactors, and RHT can use reliable extrapolations

Extensions

- To other bosonic theories in $1+1$ $\rightarrow$ easy **NEW**: done
- To fermionic theories in $1+1$ $\rightarrow$ feasible **NEW**: $\simeq$ done via bosonization
- To $2+1$ and $3+1$ dimensions $\rightarrow$ very difficult
  (lattice tensor networks will probably rule in $2+1$ and $3+1$ for numerics)

Summary

1. New ansatz for $1+1$ relativistic QFT
2. No cutoff, UV or IR (a first?)
3. UV is captured exactly even at $D = 0$
4. Efficient (cost poly D, error superpoly $1/D$)
5. Rigorous (variational)